

CS 462 – Computer Networks

Syllabus

Department of Electrical Engineering and Computer Science

Embry-Riddle Aeronautical University

Daytona Beach Campus

Fall 2026

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Office Hours: *MWF* 11am-12pm, 1-2 pm;
or by appointment.

TA:

TA Email:

TA Office Hours:

TA Office:

Course Description:

Modern computer network analysis and design using a structured approach based on the implementation of the ISO/OSI model and current standardized protocols. Network architectural layers above the physical layer - only a brief overview is presented on the physical layer.

Prerequisites: CEC 320.

Supplemental Text:

- Computer Networking: A Top-Down Approach, by Jim Kurose with Keith Ross, Pearson

Course Objectives:

The purpose of the course is to:

- Understand the different layers of the ISO/OSI model and what each layer deals with.
- Analyze and evaluate network architectures by applying the ISO/OSI model, focusing on the functionalities and interactions of network layers above the physical layer.
- Design and optimize network solutions using a structured approach that integrates current industry standards and best practices
- Understand different transmission types and the characteristics of each

Learning Outcomes

Upon completion of this course, students should be able to:

- Understand the different layers in the ISO/OSI model.
- Analyze information captured from several layers.

- Use programs to capture transmission data.
- Understand the different security flaws for each layer.

Grading:

Students will receive grade points on assignments, examinations, a course project. Each student's final course grade will be determined by a weighted sum of grade points received during the course. The relative weight of each grading element in the final grade will be computed as follows:

Assignments	30%
Class Participation	10%
Quizzes	10%
Examinations:	30%
Final Report	20%

The course letter grade is awarded as follows:

A	100 – 90
B	89 – 80
C	79 – 70
D	69 – 60
F	less than 60

Course Policies:

The examinations will be roughly equally spaced through the term and will emphasize material covered since the previous examination. Resources outside of the required textbook may be posted in Canvas or discussed in-class.

Attendance is not mandatory. However, students are highly encouraged to attend all class sessions to benefit from the in-class discussions and interactions. If a student does not attend a class, it is the student's responsibility to ensure that they determine what occurred during the class, including any of the instructor's announcements concerning assignments and due dates.

Students may be excused from deadlines or examinations only in the event of a medical or personal emergency sanctioned by the University Office of Judicial Affairs. Makeup of any examination excused on such a basis must be arranged with the instructor.

Work submitted late will be penalized by deducting a letter grade per every 24-hour period past the due date up to a 50 grade, after this threshold the late penalty is not assessed. Work not submitted will count as 0.

Academic Integrity:

Embry-Riddle Aeronautical University is committed to intellectual integrity and considers dishonesty a very serious offense. Such offenses include cheating (accepting unauthorized assistance in preparing assignments), fraud (gaining unfair advantage through deceit, trickery, or falsification of records), or plagiarism (taking the ideas, writings, words, and/or work of another and representing them as one's own without appropriate acknowledgement). A student who cheats, commits any academic fraud or plagiarizes in this course will receive an F for the course. In addition, the incident will be reported to the Dean of Students. If any other academic integrity violations have been documented, the student will be recommended for dismissal from the University.

Disability Statement:

Embry-Riddle Aeronautical University recognizes its responsibility under the mandates of Section 504 of the Rehabilitation Act of 1973 and Title III of the Americans with Disabilities Act of 1990, to provide equal access to its programs and services for students with disabilities. To assure nondiscrimination, the University is prepared to make reasonable accommodations to promote students' effective participation in their academic and co-curricular objectives. If you have a documented disability (or need to have a disability documented), and need an accommodation, please contact me as soon as possible, so that we can discuss with the Disability Support Services how to meet your specific needs and the requirements of the course. For more information, please visit the [Disability Support Services](#).

Disclaimer:

This document is NOT a contract between the students and the instructor. The instructor reserves the right to modify the contents of this syllabus. Notification of changes will be communicated to the students, and the updated document will be posted in Canvas.

CS 462 Tentative Schedule

Fall 2026

Week	Day #	Date	Topic	Deadline
1	1	8/24/26	Introduction and Syllabus	
1	2	8/26/26	Introduction	
1	3	8/28/26	Application	
2	4	8/31/26	Lab 0	
2	5	9/2/26	Application	
2	6	9/4/26	Application	
3		9/7/26	Holiday-Labor Day	
3	7	9/9/26	Transport/Hands on	Quiz Application
3	8	9/11/26	Transport	
4	9	9/14/26	Transport	Hands on due (if any)
4	10	9/16/26	Transport/Hands on	Quiz Transport
4	11	9/18/26	Network/Hands on	

5	12	9/21/26	Network	
5	13	9/23/26	Network	Hands on due (if any)
5	14	9/25/26	Network	
6	15	9/28/26	Review	Quiz Network 1
6	16	9/30/26	Exam 1	
6	17	10/2/26	Exam Revision & Final Project	
		10/6/26	Career Expo	
7	18	10/5/26	Network 2	Hands on due (if any)
7	19	10/7/26	Network 2	
7	20	10/9/26	Network 2	
8	21	10/12/26	Network 2/Hands on	
8	22	10/14/26	Network 2/Hands on	Quiz Network 2
8		10/16/26	Fall Break	
9	23	10/19/26	Link Layer	
9	24	10/21/26	Link Layer	Hands on due (if any)
9	25	10/23/26	Link Layer	
10	26	10/26/26	Wireless & Mobile	Quiz Link Layer
10	27	10/28/26	Wireless & Mobile	Hands on due (if any)
10	28	10/30/26	Wireless & Mobile	
11	29	11/2/26	Wireless & Mobile /Hands on	Quiz
11	30	11/4/26	Review	Quiz Wireless
11	31	11/6/26	Exam 2	
12	32	11/9/26	Exam Revision	Hands on due (if any)
12		11/11/26	Holiday-Veteran's Day	
12	33	11/13/26	Network Security	
13	34	11/16/26	Network Attacks	
13	35	11/18/26	Network Attacks/Hands on	
13	36	11/20/26	Emerging Technologies	
14	37	11/23/26	Troubleshooting	
14		11/25/26	Thanksgiving Break	
14		11/27/26	Thanksgiving Break	
15	38	11/30/26	Hands on/ Project Presentations	Hands on due (if any)
15	39	12/2/26	Project Presentations	Project Report